PRD — Methodica (MVP Final Spec)

Product Name: Methodica
Positioning: *Your statistical co-pilot — Graphpad's Prism, but smarter.*
Audience: Students & new researchers (bio-first, multi-field via context)
MVP Objective:

> Enable a new researcher to go from raw data → **defensible analysis + publication-ready figure + methods text** in **under 3 minutes**, while preventing silent statistical mistakes.

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0. Executive Summary (Non-Negotiable Goals)

1. **Intent-first workflow** — users start from *what they want to show*, not statistical tests.
2. **Correctness over flexibility** — no silent misuse of statistics.
3. **AI assists, never decides truth** — AI proposes; deterministic engine computes.
4. **Local compute by default** — raw data never leaves the device.
5. **Override allowed, always logged** — users can be wrong, but knowingly.
6. **Trust surfaces visible** — equations, assumptions, citations, reviewer critique.

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1. Target Users & Constraints

Primary Users (MVP)

- Undergraduate / graduate students
- New wet-lab researchers
- Minimal formal statistics training

- Goal-driven (“I want to compare treated vs control”)

Constraints

- **Solo use only** (no collaboration)
- **Web UI + local compute**
- **Field selection = contextual hint**, not a mode switch
- **Freemium** (no accounts required in MVP)

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2. Core Problems (Must Be Solved)

1. *“I can click buttons but I don’t know if it’s the right analysis.”*
2. *“I don’t know which buttons to click — just suggest the right graph/test.”*

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3. MVP Scope

3.1 Data Ingestion

- CSV upload
- XLSX upload
- Copy-paste table
- Local file reference preserved

3.2 Schema Inference

Automatically infer:

- Outcome variable (Y)
- Predictor (X)
- Grouping variable(s)
- Replicates
- Pairing (pre/post)

Ask user **only if ambiguity remains**, max **3 questions**.

3.3 Supported Analyses (MVP)

A. Two-Group Comparison

- Paired / unpaired
- Parametric (t-test)
- Non-parametric fallback (Mann-Whitney / Wilcoxon)

B. One-Way Multi-Group

- ANOVA
- Kruskal–Wallis
- Multiple comparisons:
    - Dunnett (vs control)
    - Tukey (all pairs)

C. Correlation & Regression

- Pearson / Spearman
- Linear regression
- Basic diagnostics summary

D. Dose–Response

- 4PL / 5PL
- EC50 / IC50
- Top/bottom constraints

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4. Outputs

- Publication-ready figure (default best practices)
- Statistical results table
- Explain view:
    - rationale
    - assumptions
    - equations
    - citations
    - reviewer critique
- Export formats:
    - PNG (Free)
    - SVG / PDF (Pro)
    - Methods text (TXT; DOCX optional later)

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5. Explicit Non-Scope

- Two-way ANOVA
- Mixed-effects models
- Survival analysis
- Plugins / scripting
- Collaboration
- Regulatory compliance (FDA/GxP)

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6. Product Principles → Hard Requirements

P1. Intent-First

Requirement: No analysis runs without an explicit goal step.

P2. AI Suggests, Engine Decides

Requirement:

- LLM never computes statistics
- All numbers come from deterministic engine

P3. No Silent Errors

Requirement:

- Assumption failures generate structured warnings

P4. Override With Accountability

Requirement:

- Risky actions require acknowledgment

- Logged in audit trail

P5. Trust Surfaces Visible

Requirement:

- Explain tab generated from plan + diagnostics
- Equations + citations mandatory for Pro

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7. Architecture

7.1 Two-Plane System

Control Plane (Cloud)

- Web UI hosting
- AI planner
- Citation knowledge base
- Licensing verification
- Privacy-safe telemetry

Data Plane (Local)

- Local Analysis Agent (HTTP daemon)
- Runs:
    - parsing
    - validation
    - assumption checks

    - stats
    - plot spec generation
    - exports
- Stores raw data locally

Default: Local compute ON, cloud OFF

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8. Local Analysis Agent (MVP Choice)

Packaging

- **Daemon install via `pipx`**
- Command: `methodica-agent start`
- Binds to: `127.0.0.1:7337`

Security

- One-time pairing token
- Origin allowlist
- CSRF token
- No external network access required

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9. Data Privacy & AI Policy

- Raw datasets **never sent to cloud by default**

- AI receives:
  - schema
  - summary stats
  - diagnostics
  - goal text
- Raw-data AI access = future Pro opt-in

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10. Core Domain Objects

10.1 AnalysisPlan (Canonical JSON)

```
{
  "version": "1.0",
  "goal": {
    "intent":
"compare_groups|dose_response|correlation|regression|eda",
    "primary_outcome": "Y",
    "x": "X",
    "groups": ["control", "treated"],
    "paired": false,
    "control_group": "control"
  },
  "graph": {
    "type": "scatter_summary|box|violin|line|dose_response",
    "show_points": true,
    "summary": "mean|median",
    "error_bars": "SEM|SD|CI95",
    "log_x": false,
    "log_y": false
  },
  "analysis": {
```

```
    "family":
"t_test|mann_whitney|anova|kruskal_wallis|pearson|spearman|linear_regr
ession|4pl|5pl",
    "multiple_comparisons": {
      "enabled": false,
      "method": "dunnett|tukey|sidak|holm"
    },
    "alpha": 0.05,
    "confidence_level": 0.95
  },
  "outliers": {
    "policy": "none|flag_only|remove_with_justification",
    "method": "ROUT",
    "q": 1
  },
  "context": {
    "field": "biology|psychology|chemistry|other",
    "notes": "optional"
  }
}
```

10.2 AnalysisRun (Engine Output)

- `results_json`
- `warnings_json`
- `diagnostics_json`
- `plot_spec_json` (Vega-Lite)
- `methods_text`
- `explain_bundle`

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11. UI / UX Specification

11.1 Import Dataset Screen

- Left: upload / paste
- Right: table preview + detected schema

Block progression until:

- Y identified
- groups valid
- pairing resolved

“Not sure” → conservative defaults + warnings.

11.2 Goal Picker

- Tiles + free text
- Field selector (contextual only)

Field affects language & norms, **never forces a test**.

11.3 Plan Review (Mentor-like)

Cards:

1. Recommended Graph
2. Recommended Analysis
3. Assumptions & Risks

Rules:

- No issues → “Checks passed”
- Issues → amber card with ≤3 bullets

Actions:

- Accept Plan
- Edit Plan (advanced)
- Run (disabled until validated)

Override modal:

- “I understand” checkbox
- Optional justification
- Logged

11.4 Results Screen

Tabs:

- Figure
- Stats
- Explain
- Audit

Figure Defaults

- n < 30 → show points ON
- Bar-only → warn
- CI95 preferred

Explain Tab (Trust Surfaces)

- Equations
- Citations
- Reviewer critique
- Methods paragraph

Audit Tab

- dataset hash
- plan JSON
- engine version
- override log

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12. Plot System

- **Vega-Lite**
- Deterministic, auditable
- Same spec used for render + export

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13. Computation Engine Requirements

Libraries

- pandas, numpy
- scipy, statsmodels

- scipy.optimize or lmfit
- pingouin optional (consistent usage required)

Checks

- Missingness
- Normality (Shapiro–Wilk)
- Variance (Levene / Brown–Forsythe)
- Regression diagnostics
- Multiple comparison inflation

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14. Warning ID Catalog (Partial)

ID	Severity	Meaning
NORMALITY_FAIL	warning	Normality violated
VARIANCE_FAIL	warning	Unequal variances
SMALL_N	warning	n too small for reliable inference
MULTI_COMP_RISK	warning	No correction selected
BAR_WITH_SMALL_N	info	Bar chart may obscure variability
OUTLIER_REMOVAL	critical	Outlier removal requires justification

Warnings are structured objects:

```
{
  "id": "NORMALITY_FAIL",
  "severity": "warning",
  "message": "Normality test failed in treated group.",
  "recommended_action": "Use non-parametric test.",
  "evidence": { "test": "shapiro", "p": 0.01 }
```

```
}
```

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15. APIs

15.1 Local Agent API

Base: `http://127.0.0.1:7337/v1`

- `POST /health`
- `POST /dataset/parse`
- `POST /dataset/summarize`
- `POST /plan/validate`
- `POST /run`
- `POST /export`

15.2 Cloud API

- `POST /ai/goal-spec`
- `POST /ai/propose-plans`
- `POST /ai/explain`
- `GET /kb/citations`
- `POST /license/verify`
- `GET /entitlements`

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16. AI Planner Rules

- Input: schema + summaries only
- Output: valid AnalysisPlan JSON
- Never claim assumptions are met
- Never output numbers

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17. Freemium Model

Free

- Unlimited local runs
- PNG export
- Limited Explain

Pro

- SVG/PDF export
- Full Explain
- Methods export

License verified offline via signed key.

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18. Definition of Done (MVP)

- User can produce correct analysis in < 3 min

- No silent assumption violations
- All risky actions logged
- Results reproducible across OS
- No raw data leaves device

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19. Sprint Plan

Sprint 1: Import + schema + agent skeleton
Sprint 2: Goal → Plan → Validate
Sprint 3: Core analyses + figures
Sprint 4: Dose-response + Explain + Audit
Sprint 5: Freemium + export polish